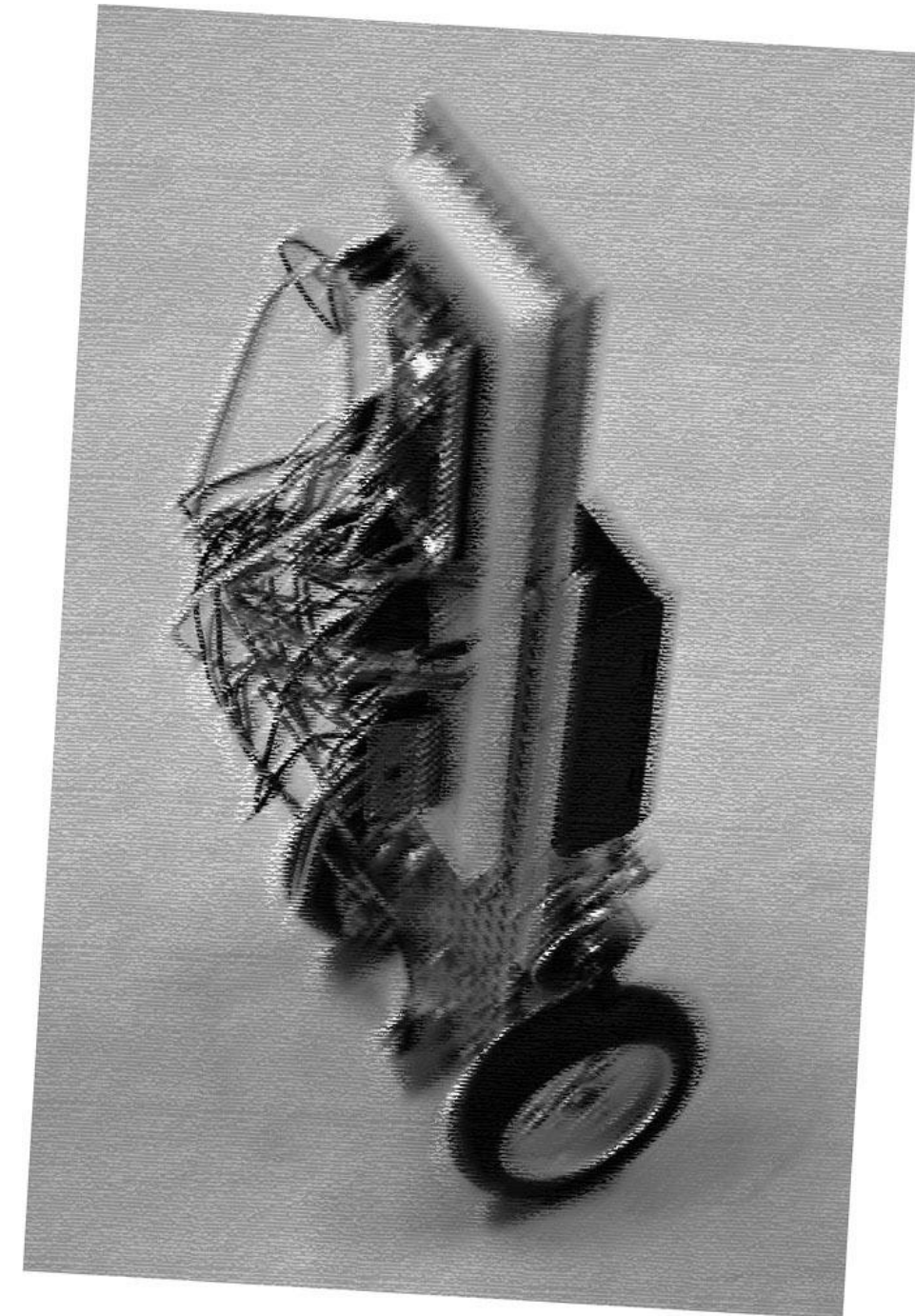


PRACTICAL EXERCISE · WEEK 1

Construction of a Wheeled Inverted Pendulum

Shaoqi (シャオ チー)

COURSE Department of Precision Engineering · 2026 S1S2
SESSIONS Mondays, 13:00–18:00 · Room 326, Engineering Bldg. 14
TODAY Course Introduction — 10 minutes



Week 0 - You Should Have Done These.

✓ Register a Keil Studio Cloud Account

✓ Install Processing 4

✓ Install Fritzing

✓ Download the Course Material

No worries if you don't get it done.

THE FOUR-WEEK ARC

Timeline

1

Week 1
Standing

Assembly & bring-up
Angle-only **standing** demo

2

Week 2
Position control

Build a rotary encoder
Position control
Simulation in Simulink

3

Week 3
EXTENSION

Load & disturbance tests
Advanced control (LQR/MPC)
.....

4

Week 4
presentation

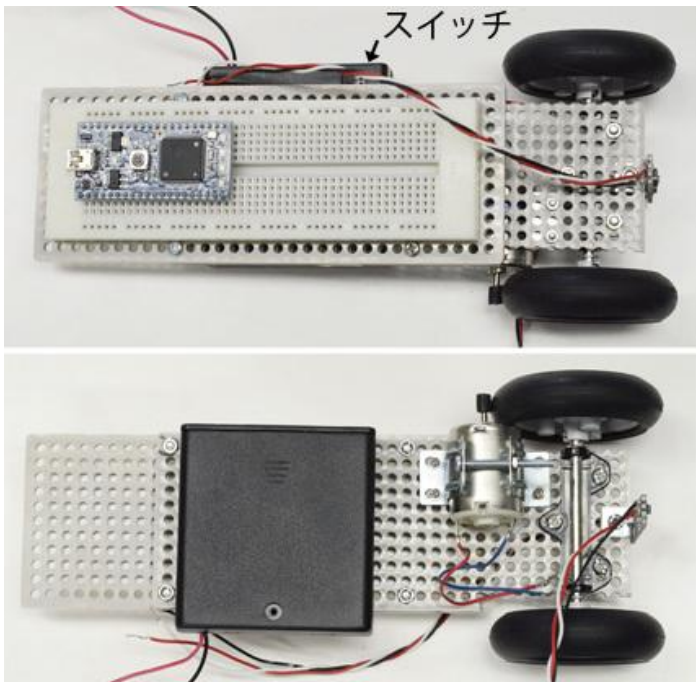
10-minute presentation
Live demo & Q&A
Final grade · 40%

Step 0 - Work in parallel.

Assign the tasks among yourselves. If there are four people in the group, two people can work on one task together. Alternatively, the fourth member can begin circuit assembly.

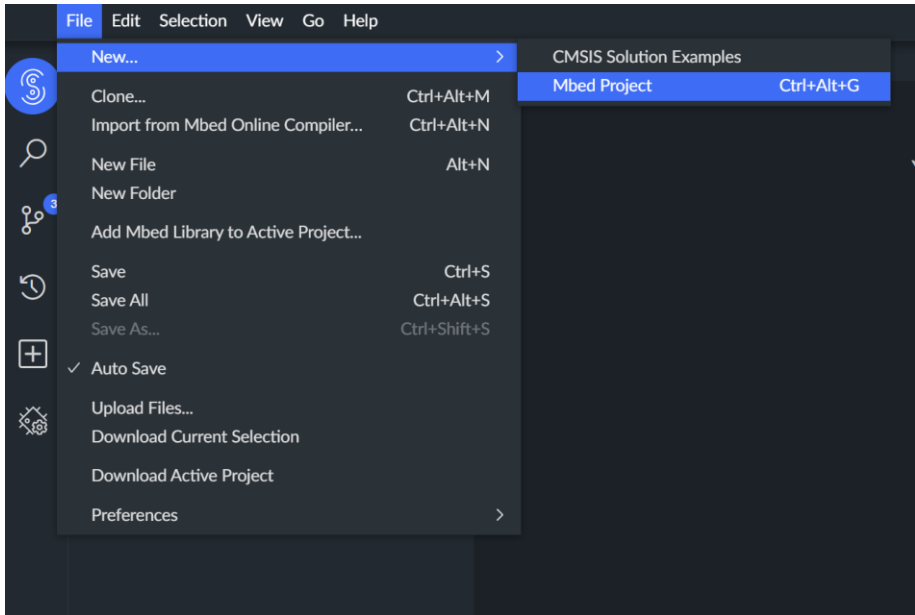
A TASK A
STUDENT A · MECHANICAL

Mechanical assembly



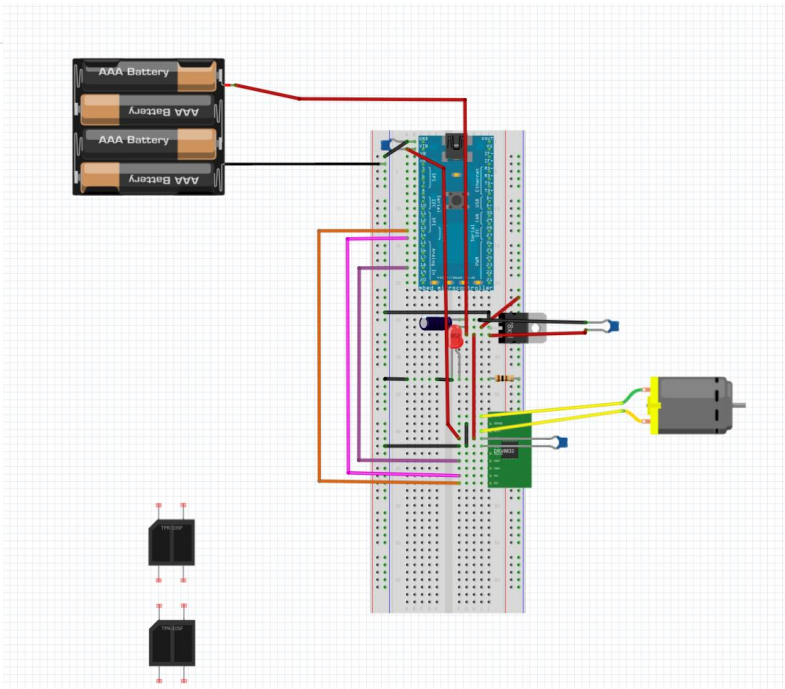
B TASK B
STUDENT B · FIRMWARE

Microcontroller



C TASK C
STUDENT C · WIRING DESIGN

Fritzing breadboard layout

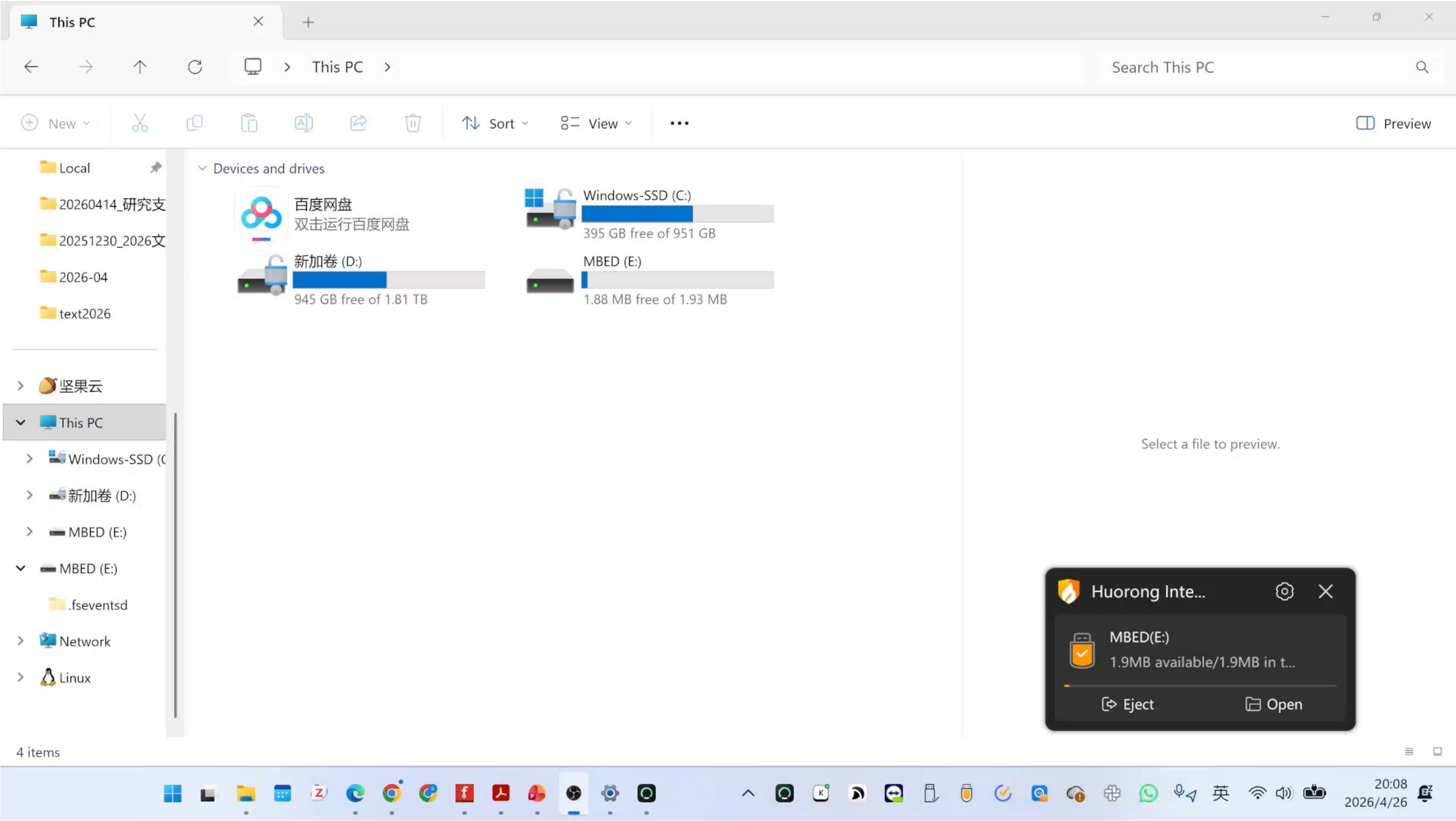


Microcontroller

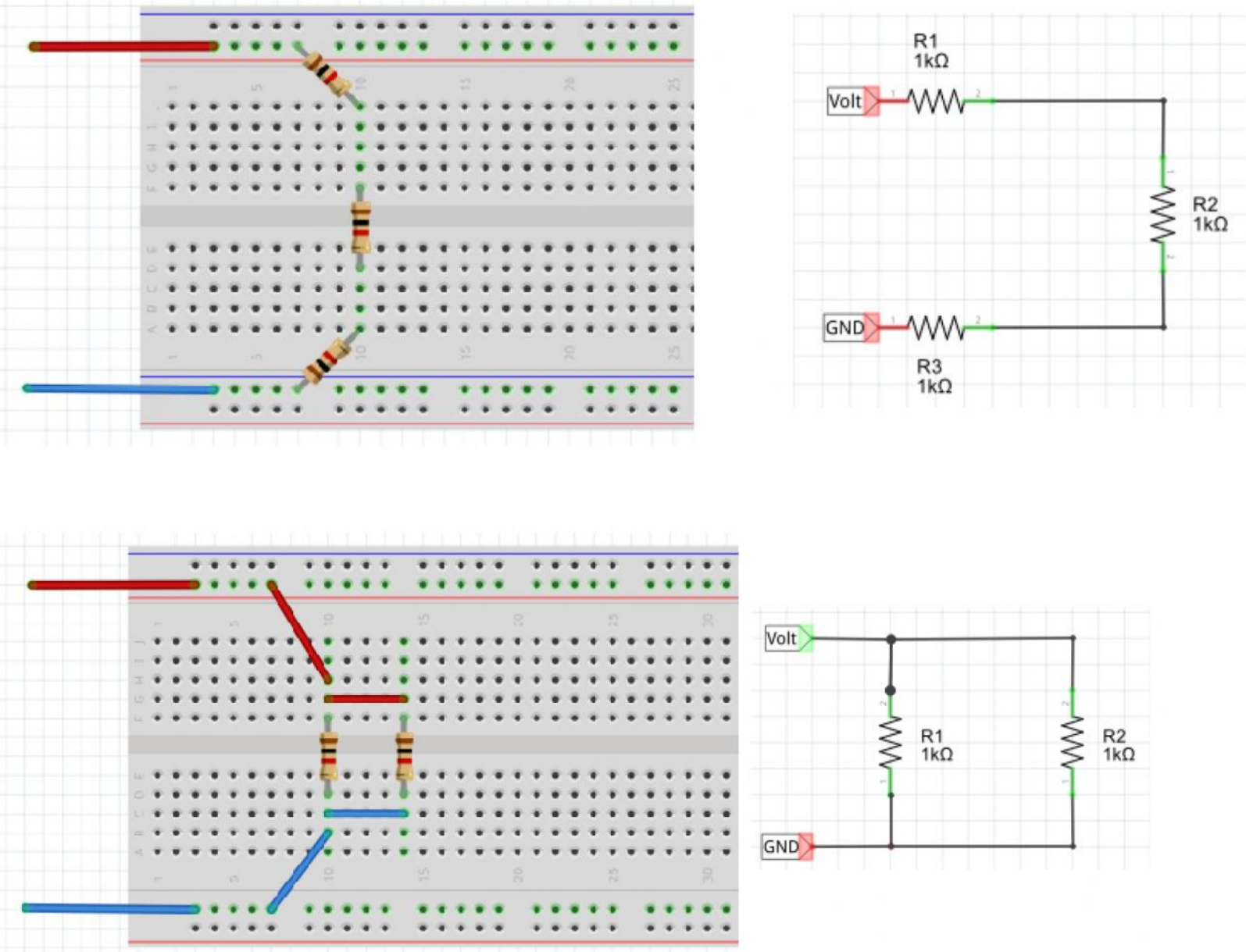
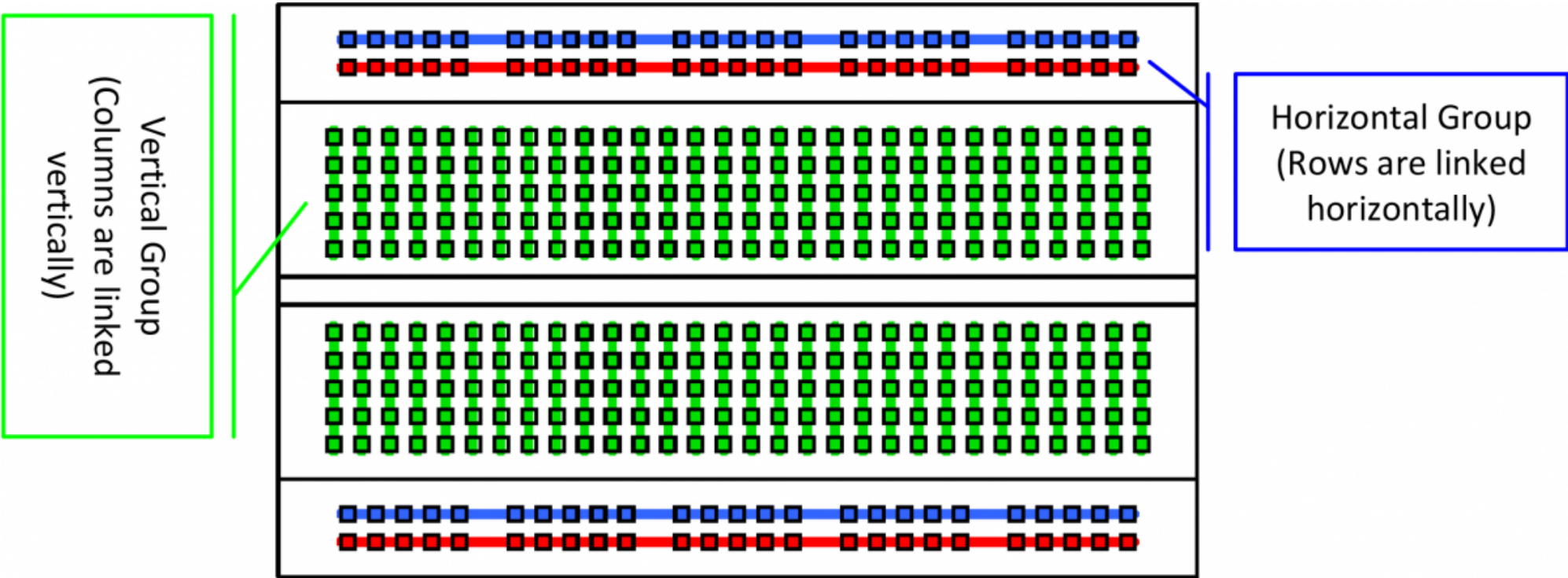
course_materials\debug_tools

-  **D1_motor_controller.bin** **compiled binary**
generated by Keil Cloud → Copy to Microcontroller
-  **D1_motor_controller.cpp** **source code**
Open with Notepad → Copy to Keil Cloud
-  **D1_motor_controller.pde** **debugging tool**

Microcontroller

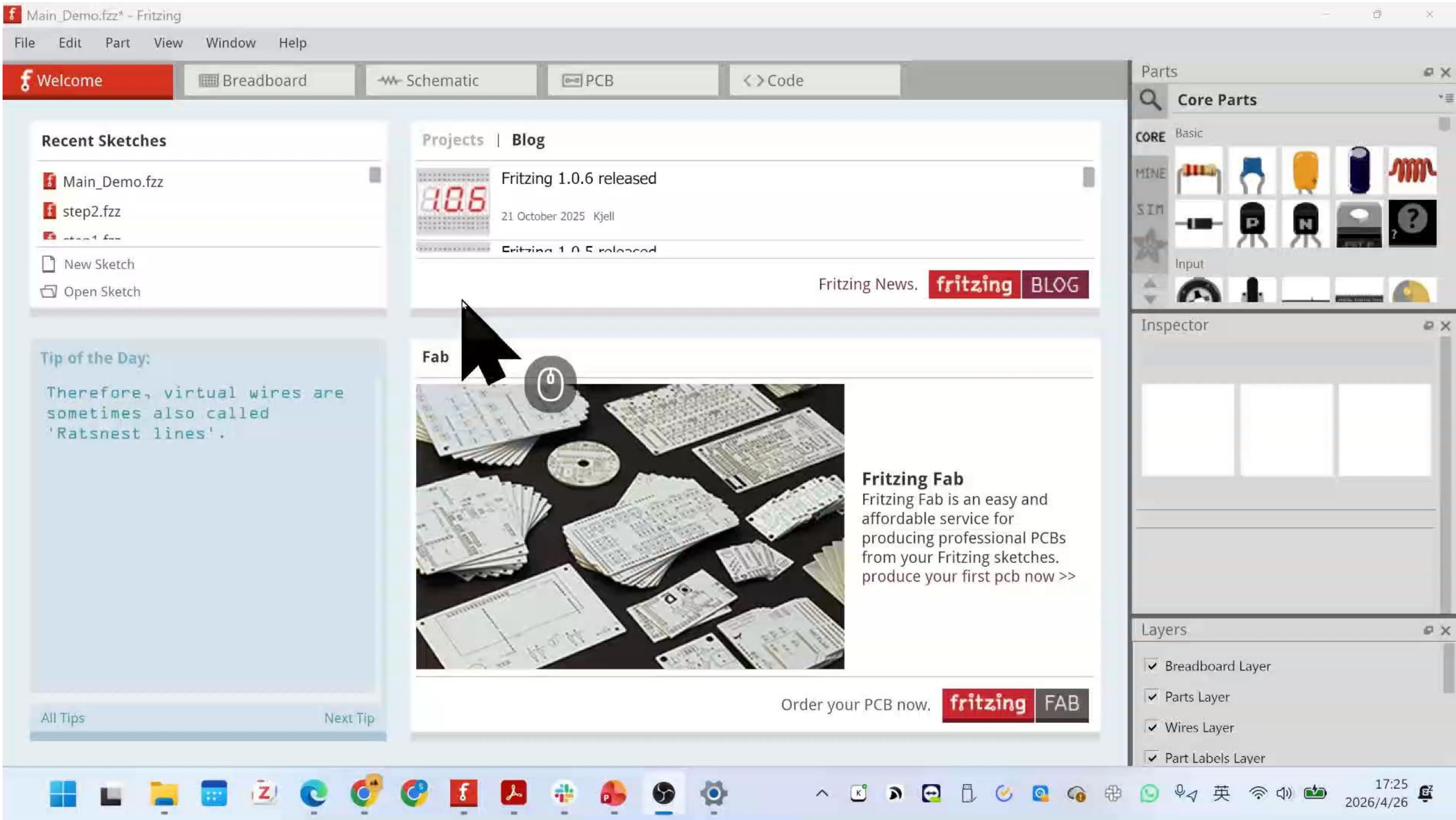


Breadboard



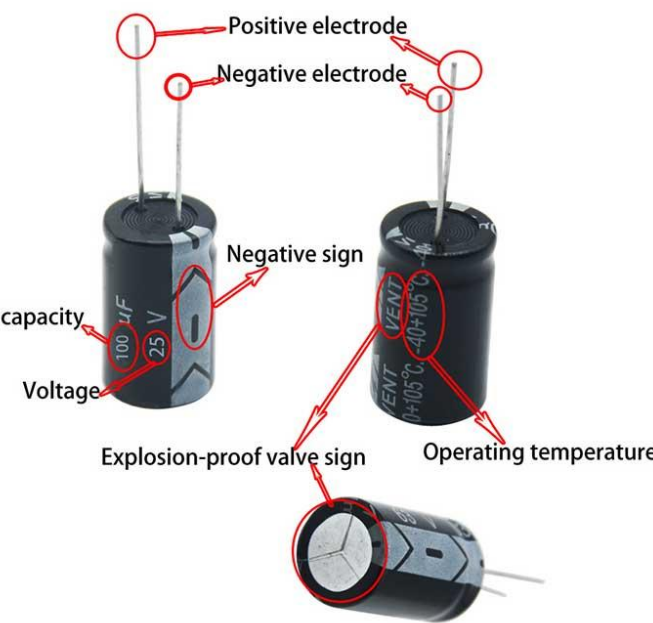
It's quite abstract.

Fritzing-assisted breadboard layout

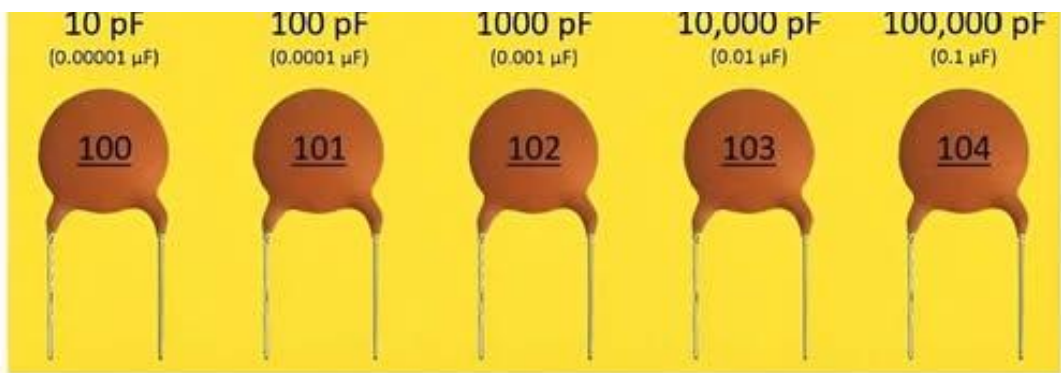


Component values and polarity

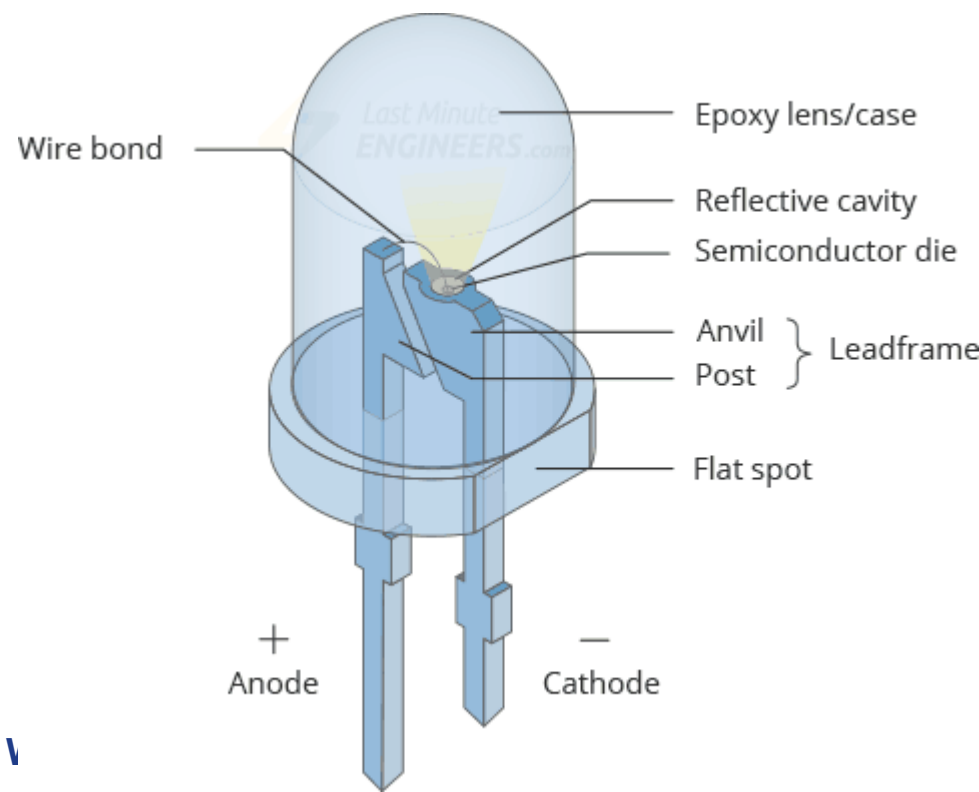
NONPOLARIZED POLARIZED



$104 = 10 \times 10^4 = 100,000 \text{ pF} = 0.1 \mu\text{F}$



0.1µF	104	100nF	100,000pF
0.12µF	124	120nF	120,000pF
0.15µF	154	150nF	150,000pF
0.22µF	224	220nF	220,000pF

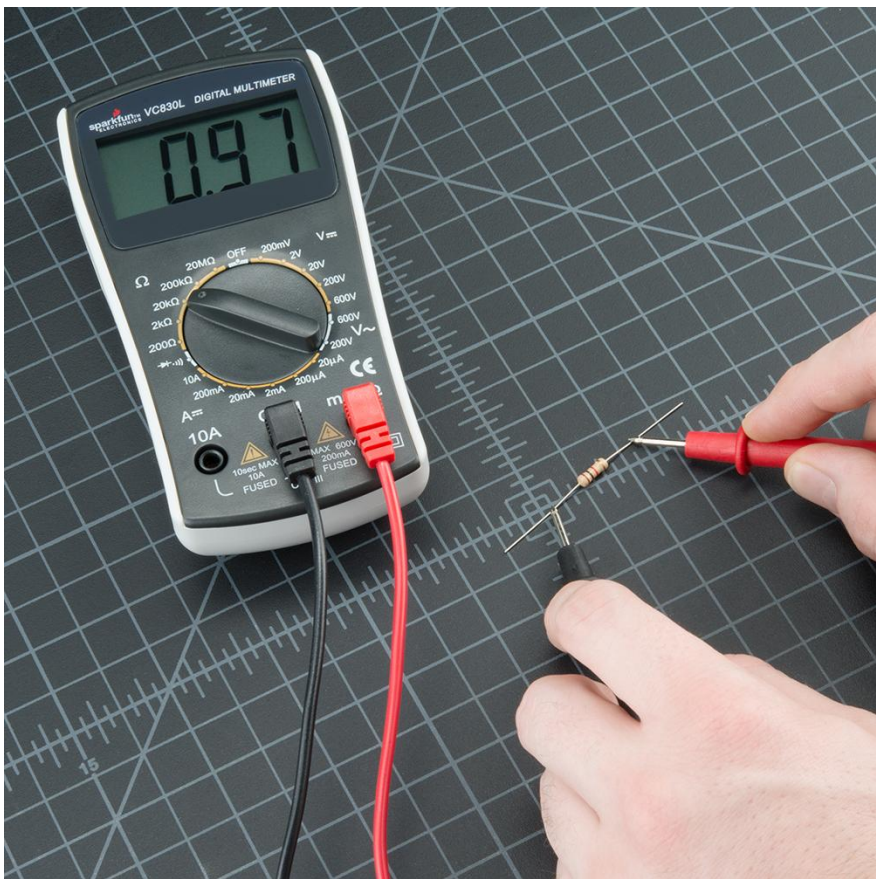


Example resistor: 1 6 0 x1k = 160kΩ

Color	1st	2nd	x th	Multiplier	Tolerance
Black	0	0	0	x1	
Brown	1	1	1	x10	±1%
Red	2	2	2	x100	±2%
Orange	3	3	3	x1k	±3%
Yellow	4	4	4	x10k	±4%
Green	5	5	5	x100k	±0.5%
Blue	6	6	6	x1M	±0.25%
Purple	7	7	7	x10M	±0.1%
Gray	8	8	8	x100M	
White	9	9	9	x1,000M	
Gold				x0.1	±5%
Silver				x0.01	±10%

Common Tolerances

Reading Axial Resistor Codes



Team Collaboration (3 STEP)

TOOL115200

motor_controller

Motor Control (mbed: F<0..100>, B<0..100>, S)

Port: (no ports) (click to cycle)RefreshConnect

Forward (F)Backward (B)Stop (S)

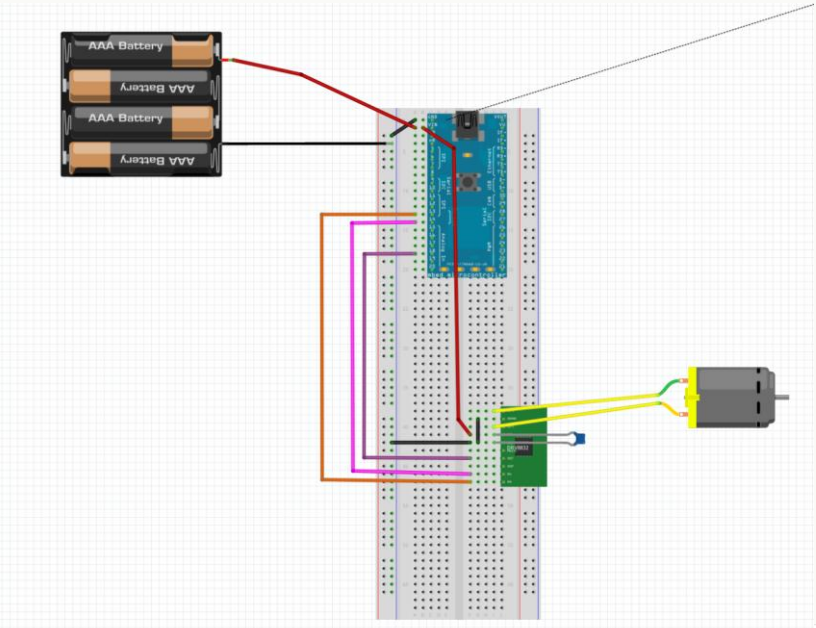
Duty (%): 30

Drag sets duty. On mouse release, sends F/B <duty> if motor is running.

Expected IO State (UI-side)
DIR = STOP
p13(motor_f) = 0p14(motor_r) = 0p18(motor_v duty) = 30%
Note: shown state is what UI sent; mbed does not send real IO feedback.

Log: Ready.

STEP 1 Motor Controller



TOOL115200

analog_input_visualizer

4ch Analog Debug (Frame: AA 55 CH1 CH2 CH3 CH4)

Port: COM3 (click to cycle)RefreshDisconnect

Decoded Channels (raw 0..255) [CH1=p16, CH2=p17, CH3=p19, CH4=p20]
CH1 (p16): 79CH2 (p17): 62CH3 (p19): 151CH4 (p20): 10
CH1: 0.310CH2: 0.243CH3: 0.592CH4: 0.039
Frames: 897 Bad bytes: 0 (auto re-sync by header)

Simple Bar View (0..1)
CH1 p160.310
CH2 p170.243
CH3 p190.592
CH4 p200.039

Tip: if values look frozen -> check baud / wiring / device is sending frames.
Log: Connected: COM3 @ 115200

STEP 2 Analog Input Visualizer

TOOL230400

six_channel_oscilloscope

Serial Port: COM3ConnectPauseArm Save

Status: Connected | Running | Save -- (Port: COM3)

CH1: p16 (0..1)
CH2: p17 (0..1)
CH3: p19 (0..1)
CH4: p20 (0..1)

CH1: omega
CH2: omega
CH3: id
CH4: id

STEP 3 · Standing Example

AI tools may be used for editing and refinement.

Step 1 – Motor Controller

TOOL115200

motor_controller

Motor Control (mbed: F<0..100>, B<0..100>, S)

Port: (no ports) (click to cycle)

RefreshConnect

Forward (F)Backward (B)Stop (S)

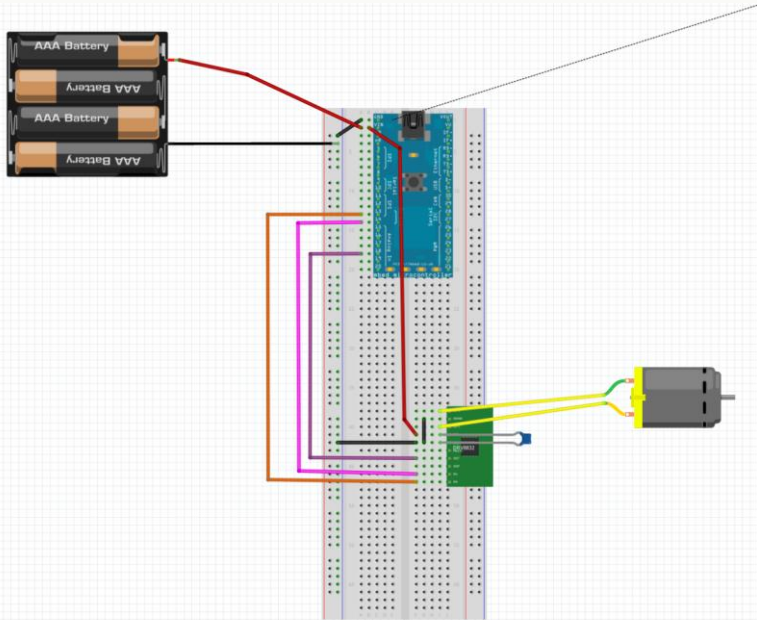
Duty (%): 30

Drag sets duty. On mouse release, sends F/B <duty> if motor is running.

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Note: shown state is what UI sent; mbed does not send real IO feedback.

Log: Ready.

STEP 1 Motor Controller



D1_motor_controller

← → ↑ ↺ ↻ > ... text2026 > course_materials > debug_tools > D1_motor_controller

Search D1_motor_controller

+ New ✂️ 📄 📁 📄 📄 🗑️ ↕️ Sort ▾ ≡ View ▾ ⋮

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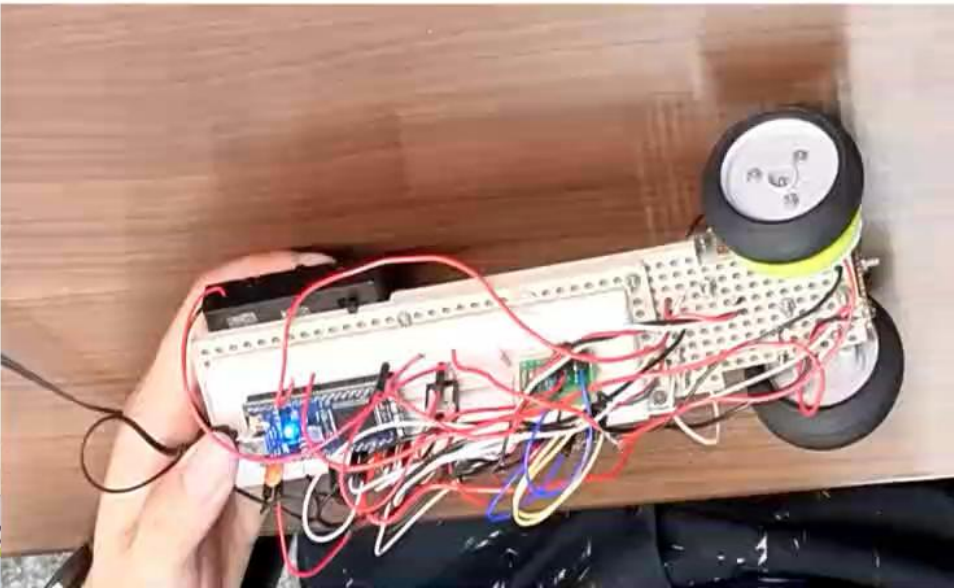
📁 坚果云 This PC

fritzing and 3 more tabs -... D1_motor_controller... Downloads - File Explorer MBED (E:) - File

5 items 1 item selected 1.75 KB

Name	Date modified	Type	Size
D1_motor_controller.bin	2026/1/3 16:45	BIN File	28 KB
D1_motor_controller.cpp	2026/1/3 16:49	CPP File	2 KB
D1_motor_controller.pde	2026/1/3 17:05	PDE File	8 KB
README.md	2026/2/4 13:05	Markdown File	7 KB
README.pdf	2026/2/4 13:05	Adobe Acrobat 文...	49 KB

The file you are attempting to preview could harm your computer. If you trust the file and the source you received it from, open it to view its contents.



Grading Criteria

Weekly submissions · 60%

Final Presentation · 40%

Week	Deliverables	Score (%)	Notes
W1	— One video: standing demo	60% (Weeks 1–3 submissions)	Due by 9 pm on the day of the session. Late until 9 pm the following day ; the maximum score is 50% . → See Week 1 submission details (p. 24)
W2	— Video 1: real hardware — Video 2: simulation without disturbance — Video 3: simulation with disturbance		Due by 9 pm on the day of the session. Late until 9 pm the following day ; the maximum score is 50% . → See Week 2 submission details (p. 43)
W3	— Video 1: maximum disturbance/load on real hardware — Video 2: maximum disturbance in simulation — One-page PDF summary		Due by 9 pm on the day of the session. Late until 9 pm the following day ; the maximum score is 50% . → See Week 3 submission details (p. 45)
W4	Final presentation & Q&A	40%	Live 10-minute group presentation; no written submission required. → See Week 4 details (p. 47)

DEADLINE

21:00

on the day of the class

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Presentation in English

Week 1 Submission

Video (max 200 MB):
show the wheeled inverted pendulum
balancing upright for at least 5 seconds.

Format MP4 · smartphone recording recommended

Submit to forms.gle/5huiVLJAj5Bh16Dj7 [Week1_GroupX.zip](#)

If the robot cannot yet stand, record its current state and clearly show which parts are already working.

Each group only needs one submission.

DEADLINE

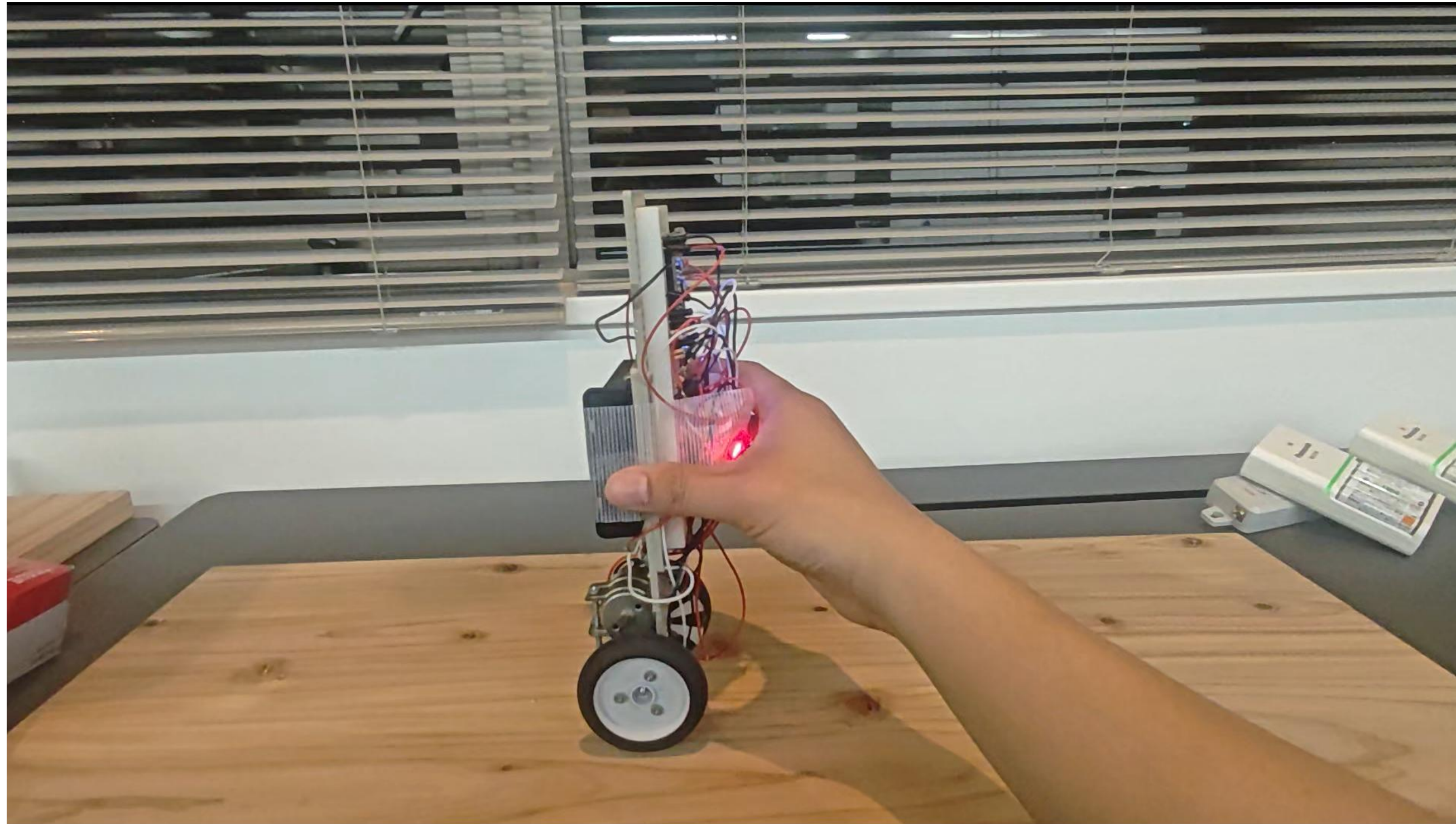
21:00

on the day of the class

Today — Monday, April 27

Late submissions accepted until **21:00 the following day** . Late score capped at 50%.

Week 1 Submission



Reminders

- If any parts are missing from the kit, please contact the TA.
- Some components in the kit may be damaged, so please test them by yourself.
- After flashing the .bin file, please restart the MCU.
- Remember to charge the battery.

UTokyo-Control-Practice-2026 / fritzing /

UTokyo2026 Add Fritzing demo files and guide videos (4-6); rem

Name

..

DRV8832.fzpz

Main.fzz

Main_Demo.fzz

Step1.fzz

Step2.fzz

TPR-105F.fzpz

mbed.fzpz

UTokyo-Control-Practice-2026 / video /

UTokyo2026 Add Fritzing demo files and guide videos (4-

Name

..

1_mechanical_assembly.mp4

2_circuit_assembly.mp4

3_debugging_tuning.mp4

4_keil_cloud_guide.mp4

5_fritzing_guide.mp4

6_debug_tool_guide.mp4